

	Software Engineering										Course Type			LTP		
Course Code:	CSE3005										Credits			4		
Prerequisite:																
Course Objectives: 1. To discuss concepts related to software engineering life cycle. 2. To describe about the concepts of software project estimation and software designing. 3. To explain about the concepts of software testing. 4. To discuss about an overview of object oriented analysis and design, modeling language. 5. To discuss about software maintenance and quality management.																
Course Outcomes (CO): Students will be able to CO1. Analyze and identify a suitable software development life cycle model for an application [KL4] CO2. Develop software requirements specification and cost estimation for an application. [KL4] CO3. Construct the design models and testing techniques for implementing software. [KL3] CO4. Apply the object-oriented concepts in software development. [KL4] CO5. Implement the software development process for managing software quality and maintenance using various activities from requirements to validation and verification. [KL3]																
Correlation of COs with POs																
CO \ PO	CKL	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
PKL		3	5	6	5	6	3	3	3	NA	M	3	M	3	3	3
CO1	4	3	2	1	2	1	3	3	3	2	2	3	3	3	3	2
CO2	4	3	2	1	2	1	3	3	3	2	2	3	3	3	2	3
CO3	3	3	2	2	2	2	3	3	3	2	2	3	3	3	2	2
CO4	3	3	2	2	2	2	3	3	3	2	2	3	3	3	3	2
CO5	3	3	2	2	2	2	3	3	3	2	2	3	3	3	3	3
CO	Topics to be discussed														Hrs	
CO1	SOFTWARE PROCESS MODELS The Nature of Software-A generic view of process– A layered Technology – A Process Framework– Product and Process –Software Process Models-Waterfall Model-Incremental Process Models – Evolutionary Process Models- Prototyping-Spiral Model-The RAD Model- Concurrent Model-The Concurrent Development Model – Specialized Process Models – the Unified Process- Introduction to Agile Process, Agile Management Practices, Risk Management and the Customer in Agile Methods, Agile Engineering Practices, Tailoring and Improving Agile Methods, Miscellaneous Agile Methods, Challenges in Adopting Agile Methods.														12	

CO2	REQUIREMENT ENGINEERING Requirements Engineering - Establishing the Groundwork - Eliciting Requirements - Requirements Analysis –SRS Documentation- Metrics in the Process and Project Domains – Software Measurements - Software Project Estimation - Decomposition Techniques-Empirical Estimation Models - The Make/Buy Decision.	09
CO3	DESIGN CONCEPTS AND TESTING The Design Concepts - The Design Model - Architectural Design –Structured Analysis (DFD model) – Structured Design (Structural Chart)- User Interface Design: Interface Analysis - Interface Design Steps - Requirements Modeling - Software Testing Fundamentals – Black Box Testing - White Box Testing - Unit Testing - Integration Testing -Validation testing –system testing –Art of debugging	10
CO4	OBJECT ORIENTED LIFE CYCLE Overview of Object-Oriented Systems Development - Object Oriented System Development Methodology - Unified Approach - Object Basics - Object Oriented Systems Development Life Cycle. -Object Oriented Methodologies - Rumbaugh Methodology - Booch Methodology – Jacobson Methodology Unified Approach.	09
CO5	QUALITY & MAINTENANCE Software Maintenance -Critical Systems Validation -Quality Management -Process Improvement -Risk Management- Configuration Management	06
	Total Lectures:	45
List of Experiments: 1. Course Registration System 2. Student marks analyzing system 3. Railway domain 4. Online ticket reservation system 5. Platform assignment system for the trains in a railway station 6. Medicine domain 7. Expert system to prescribe the medicines for the given symptoms 8. Remote computer monitoring 9. Finance domain 10. ATM system 11. Stock maintenance 12. Human Resource management 13. Quiz System 14. E-mail Client system.		
Text books:		
1. Roger S.Pressman, Software Engineering: A Practitioner’s Approach, McGraw Hill International edition, Seventh edition, 2009. 2. Ian Sommerville, Software Engineering, 8th Edition, Pearson Education, 2008.		
Reference Books:		

1. Stephan Schach, Software Engineering, Tata McGraw Hill, 2007
2. Pfleeger and Lawrence Software Engineering: Theory and Practice, Pearson Education, second edition, 2001
3. Ali Bahrami, Object Oriented Systems Development, Tata McGraw-Hill, 2010
4. Ian Sommerville, Software Engineering, 8th Edition, Pearson Education, 2008.
5. Stephan Schach, Software Engineering, Tata McGraw Hill, 2007

<i>by the Board of Studies on</i>	<i>23.03.2023</i>
<i>Approval by Academic council on:</i>	<i>July 18, 2018</i>
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